

47 Cancer of the Testis



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INTRODUCTION

Testicular cancers are the most common malignant neoplasm affecting men ages 15 years to 35 years. Approximately 90% of testicular cancers are germ cell tumors, which can be broadly classified as seminoma or nonseminomatous germ cell tumors (NSGCTs). Male extragonadal germ cell tumors are also discussed in this chapter. Analogous germ cell and sex cord stromal tumors in women are not included here, but in many cases, the male and female counterparts share a common biology and therapeutic approach.

INCIDENCE AND EPIDEMIOLOGY

In the United States in 2021, the estimated incidence of testicular cancer was 9,470 cases with 440 associated deaths.¹ The cancer-specific survival rate of approximately 95% has led to millions of testicular cancer survivors around the world.² The incidence varies among geographic areas; it is highest in Scandinavia, Switzerland, and Germany and lowest in Africa and Asia.³ Although testicular germ cell tumors are most common among adolescents and young adult men, they can potentially occur in boys and men of any age or genetic background. For reasons that have yet to be discovered, the incidence of testicular germ cell tumors has increased over the past 30 years in the United States and throughout much of Europe.⁴ Furthermore, although there has been an unexplained increase in incidence in higher median income countries, the mortality rate has stabilized or improved with refinement of therapy. This improvement in outcomes for patients with germ cell tumors in Western countries contrasts with countries with lower median incomes, where the mortality rate has remained elevated.⁵

Risk Factors

Men whose family history includes a first-degree relative with testicular germ cell tumor or who have a personal history of cryptorchidism are at increased risk of developing germ cell tumors.⁶ Testicular cancer survivors are also at increased risk of a second primary cancer in the contralateral testicle. These risk factors suggest a genetic or developmental etiology for testicular germ cell tumors. Although no definitive postnatal environmental risk factors have yet been identified, a study from Sweden found a correlation between heavy cannabis use and the risk of developing testicular cancer.⁷

In the United States, testicular cancer is rare among African Americans and most common among white men.⁸ This racial disparity in testicular cancer diagnosis and its geographic distribution suggest a genetic linkage with the white phenotype, but no high-penetrance allele has yet been identified.⁴ Testicular germ cell tumors are commonly accompanied by intratubular germ cell neoplasia (ITGCN), which was reclassified by the World Health Organization (WHO) as germ cell neoplasia in situ (GCNIS) in postpubertal men.⁹ It is thought that all adult germ cell tumors, with the exception of spermatocytic tumors previously referred to as spermatocytic seminomas, arise from GCNIS.¹⁰ The widely

accepted theory is that GCNIS begins in utero.¹¹ The multifocality of GCNIS suggests a field effect within the testicle and provides a mechanistic explanation for cases of bilateral testicular cancer.

Men without a history of testicular cancer are occasionally found to have GCNIS on testicular biopsy or orchiectomy that was performed for other reasons. The incidental finding of testicular microlithiasis on ultrasound may also provide evidence of GCNIS in an otherwise healthy man.¹² The risk of testicular cancer for such individuals has not been determined, but they should be counseled regarding a potential increased risk and to report any new testicular symptoms. In men with unilateral germ cell tumors who are found to have contralateral GCNIS, the 5-year risk of development of germ cell tumor in the contralateral testicle has been historically reported to be 50%.¹³

INITIAL PRESENTATION AND MANAGEMENT

Symptoms and Signs

Testicular swelling is the most common presenting complaint, and in most cases, this symptom is discovered by the patient himself. Patients may also report a pressure-like sensation, heaviness, or mild to moderate testicular pain. This is sometimes confused with orchitis or epididymitis and treated initially with antibiotics. Acute or severe pain is rarely caused by testicular cancer and suggests a different cause such as testicular torsion. Testicular ultrasound examination should be obtained as soon as a neoplasm is suspected, and one or more hypoechoic lesions are seen within the testicle. Ultrasound distinguishes a solid from cystic mass and intratesticular from intrascrotal or extratesticular location. A solid intratesticular mass on ultrasonography is presumed to be a neoplasm and is an indication for radical inguinal orchiectomy.¹⁴

Approximately 75% of patients present with a testicular mass and no clinical evidence of metastasis (clinical stage I). Others present with metastatic disease, which may also be symptomatic. Primary tumors can be small and asymptomatic, even in the presence of metastatic disease, and occasionally they “burn out,” leaving only a fibrotic scar (burned-out primary). Metastases can be the source of clinical symptoms on presentation in such cases, and symptoms may include back pain, shortness of breath, cough, gynecomastia, hemoptysis, and weight loss.¹⁵ Ultrasonography of the testicles is helpful in establishing the diagnosis even if there is no palpable testicular mass.¹⁴ The detection of elevated serum tumor markers can also be helpful when an occult testicular primary or extragonadal germ cell tumor is suspected.¹⁶

Diagnosis

Radical inguinal orchiectomy is the standard diagnostic and therapeutic procedure for a solid intratesticular mass.¹⁷ Transscrotal orchiectomy or biopsy is specifically contraindicated because of the risk of tumor cell seeding of the inguinal and pelvic lymphatic

TABLE 47.1

World Health Organization Histologic Classification of Testicular Germ Cell Tumors**Germ cell tumors**

- Intratubular germ cell neoplasia, unclassified
- Other types

Tumors of one histologic type (pure forms)

- Seminoma
 - Seminoma with syncytiotrophoblastic cells
- Spermatocytic seminoma
 - Spermatocytic seminoma with sarcoma
- Embryonal carcinoma
- Yolk sac tumor
- Trophoblastic tumors
 - Choriocarcinoma
 - Trophoblastic neoplasms other than choriocarcinoma
- Teratoma
 - Dermoid cyst
 - Monodermal teratoma
 - Teratoma with somatic-type malignancies

Tumors of more than one histologic type (mixed forms)

- Mixed embryonal carcinoma and teratoma
- Mixed teratoma and seminoma
- Choriocarcinoma and teratoma/embryonal carcinoma
- Others

With permission from Eble JN, Sauter G, Epstein JI, et al., eds. *World Health Organization Classification of Tumours. Pathology and Genetics of Tumours of the Urinary System and Male Genital Organs*. IARC Press; 2004.

drainage. Biopsy is also of limited value because testicular germ cell tumors are heterogeneous. Removal of the entire organ is necessary to properly identify the histologic type(s) present and to select the appropriate therapy. It is reasonable to perform needle biopsy of a metastatic site in cases of occult testicular primary, burned-out primary, or extragonadal germ cell tumor, although the results of needle biopsy must always be interpreted with caution because of sampling error. The pattern of serum tumor marker elevation is also informative about the likely cell types present (seminoma or nonseminoma), as discussed in the next section.¹⁶

HISTOLOGY

Male germ cell tumors are broadly classified as seminoma and NSGCT (Table 47.1)^{9,18} Patients with pure seminoma are managed differently than patients with NSGCT or mixed histology tumors, although mixed tumors may have a component of seminoma. In this sense, when we refer to seminoma as a clinical diagnosis, it is meant as pure seminoma, whereas seminoma as a histologic pattern may also be present in mixed NSGCT.

Seminoma

The microscopic appearance of seminoma is characterized by sheets of neoplastic cells with abundant cytoplasm, round hyperchromatic nuclei, and prominent nucleoli (Fig. 47.1). A prominent lymphocytic infiltrate is common such that it is sometimes

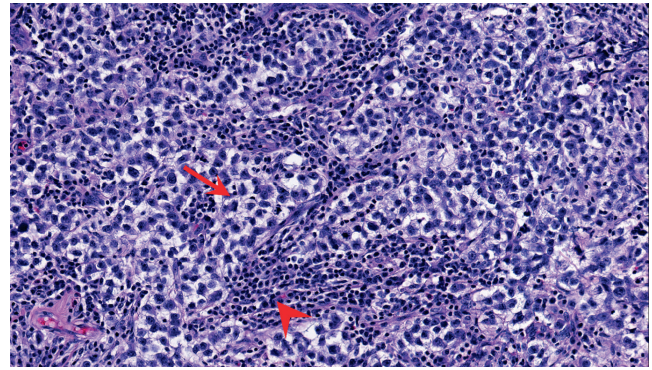


Figure 47.1 Seminoma, classic type, with neoplastic cells (arrow) and the typical accompanying lymphocytic infiltrate (arrowhead). (Reprinted with permission from Rao B, Pagliaro LC. Testicular germ cell tumors. In: Silverman PM, ed. *Oncologic Imaging: A Multidisciplinary Approach*. Elsevier; 2012:335–357.)

confused with lymphoma until the surface immunophenotype has been determined. Most seminomas do not produce serum tumor markers, but the presence of syncytiotrophoblastic giant cells in a minority of cases accounts for modest elevations of serum human chorionic gonadotropin (hCG).¹⁹ Seminomas never produce α -fetoprotein (AFP), and patients whose tumors have the histologic appearance of seminoma and whose serum AFP is elevated should be considered to have a mixed NSGCT, even if a nonseminomatous histologic pattern cannot be identified.¹⁴ Exceptions are cases in which there is another explanation for the elevated AFP, such as liver disease or a chronic nonspecific elevation.

Immunohistochemistry is usually positive for placental alkaline phosphatase (PLAP); negative for CD30, AFP, and epithelial membrane antigen (EMA); and either negative or weakly or focally positive for cytokeratin.¹⁵

Histologic variations of seminoma such as “anaplastic” or “atypical” seminoma are of no known clinical relevance. The previous entity of spermatocytic seminoma, now referred to as spermatocytic tumor by the WHO pathology classification system, has a different natural history and is even of uncertain relation to other germ cell tumors.⁹ Spermatocytic tumors usually occur in older individuals and have low metastatic potential.²⁰ Orchiectomy is the only treatment required. Unlike all other germ cell tumors, spermatocytic tumors are not associated with GCNIS.

Nonseminomatous Histologies

Germ cell tumors may be composed of a single histology or multiple histologic patterns. Through poorly understood processes of mutation, epigenetic modification, and differentiation, a single clone beginning as GCNIS can develop into an undifferentiated neoplasm (seminoma) or a primitive zygotic neoplasm (embryonal carcinoma).²¹ Further differentiation from embryonal carcinoma results in somatically differentiated tumors (teratoma) or extraembryonal differentiated tumors (yolk sac and choriocarcinoma).²² Nonseminomatous histologies are found in 55% of germ cell tumors. Male germ cell tumors that contain any histologic cell type other than seminoma or syncytiotrophoblasts are collectively referred to as NSGCT.

Embryonal Carcinoma

Embryonal carcinoma is the most undifferentiated type of germ cell tumor and is thought to be pluripotent. Cells are characterized by indistinct borders and scant cytoplasm, which can be arranged in solid sheets or in glandular or tubular structures (Fig. 47.2). On immunohistochemical staining, embryonal carcinoma can

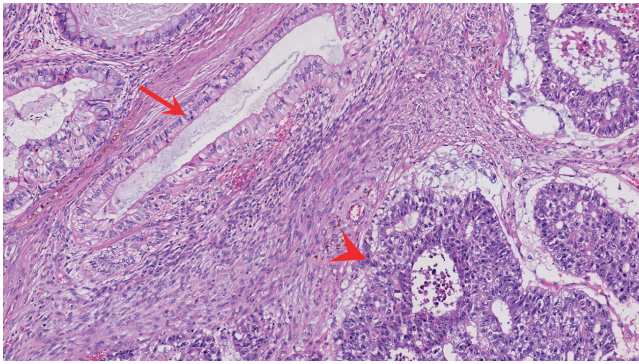


Figure 47.2 Mixed nonseminomatous germ cell tumor composed of teratoma (arrow) and embryonal carcinoma (arrowhead). (Reprinted with permission from Rao B, Pagliaro LC. Testicular germ cell tumors. In: Silverman PM, ed. *Oncologic Imaging: A Multidisciplinary Approach*. Elsevier; 2012:335–357.)

be positive for cytokeratin, CD30, PLAP, AFP, and hCG. Modest elevations of serum AFP and/or hCG can be seen, and frequently, embryonal carcinoma is marker negative. Lactate dehydrogenase (LDH) concentration in the serum is an important prognostic factor in metastatic embryonal carcinoma that is marker negative.

Choriocarcinoma

Choriocarcinoma is composed of both cytotrophoblasts and syncytiotrophoblasts.²³ The cells strongly express hCG. The clinical presentation of a choriocarcinoma-predominant or pure choriocarcinoma tumor often includes very high serum hCG levels, widespread hematogenous metastases, and tumor hemorrhage (Fig. 47.3). Syncytiotrophoblasts and syncytiotrophoblastic giant cells can be associated with other germ cell histologies, so the presence of cytotrophoblasts is required for the diagnosis.

Yolk Sac Tumor

Yolk sac tumor (endodermal sinus tumor) is commonly seen as a component of mixed NSGCT. Pure yolk sac tumors represent a significant proportion of mediastinal germ cell tumors but are rarely seen in adult testicular cancer. Histologic patterns include papillary, solid, glandular, hepatoid, macrocystic, and microcystic types. Perivascular arrangements of epithelial cells can be seen in yolk sac tumor and are known as glomeruloid or Schiller-Duval bodies. Immunostaining is diffusely positive for AFP and may also be positive for cytokeratin, SALL4, glipican-3, PLAP, and CD117. Yolk sac tumor is associated with high serum levels of AFP.

Teratoma

Teratoma arises from a pluripotent malignant precursor (embryonal carcinoma or yolk sac tumor) and contains somatic cells from at least two germ cell layers (ectoderm, endoderm, or mesoderm). Teratoma is commonly seen as a component of adult mixed NSGCT (see Fig. 47.2). A small percentage (2%–3%) of postpubertal male germ cell tumors appear to have teratoma as the only histologic type, but these are always assumed to harbor a minor component of pluripotent NSGCT and are treated the same as a mixed NSGCT.

Immature teratoma shows partial somatic differentiation, whereas mature teratoma has terminally differentiated tissues, such as cartilage, skeletal muscle, or nerve tissue and frequently forms cystic structures. Although these cells can resemble normal tissues, teratoma is a low-grade malignancy and, if untreated, grows until it is unresectable. Moreover, teratomas can give rise to secondary somatic malignancy, such as rhabdomyosarcoma, poorly differentiated carcinoma, or primitive neuroectodermal tumor.²⁴ These typically display the biology of their de novo counterparts and are treated accordingly.²⁵

Teratoma does not produce elevated serum AFP or hCG. Patients with elevated serum AFP or hCG should be assumed to have a nonteratoma germ cell tumor component unless the elevation can be otherwise explained.

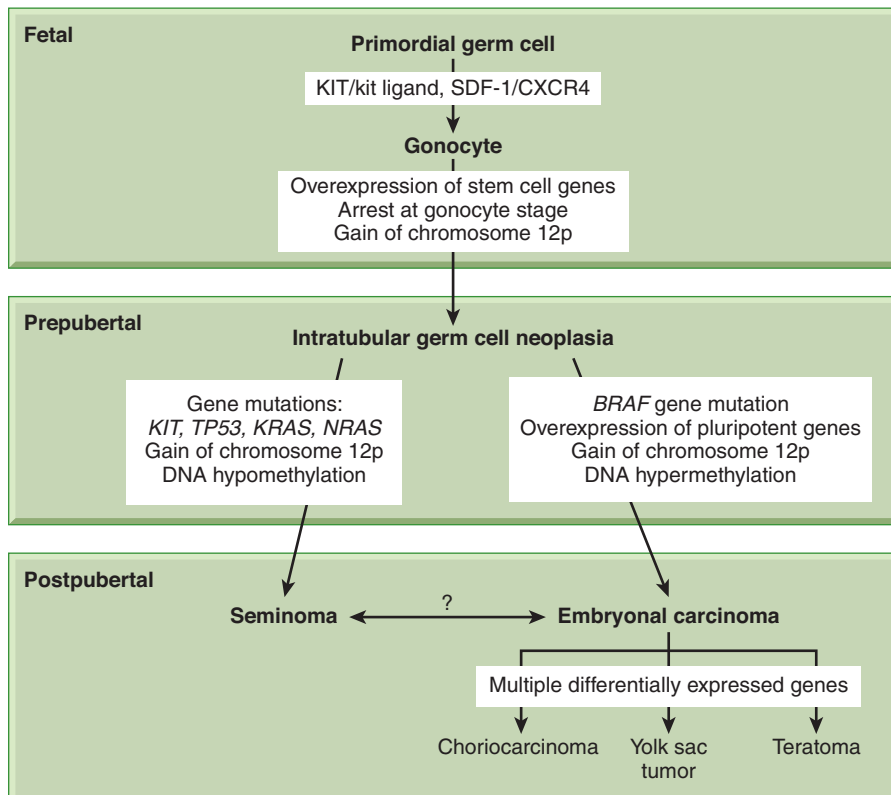


Figure 47.3 Pathogenesis of testicular germ cell tumors. ? refers to still unclear drivers of differentiator between seminomas and non-seminomas after puberty. CXCR4, C-X-C motif chemokine receptor 4; SDF, stromal cell-derived factor 1. (Adapted from Sheikine Y, Genega E, Melamed J, et al. Molecular genetics of testicular germ cell tumors. *Am J Cancer Res* 2012;2[2]:153–167.)

BIOLOGY

Mechanism of Germ Cell Transformation

Intratubular germ cell neoplasia is thought to derive from malignant transformation of primordial germ cells or gonocytes during fetal development.¹⁰ Primordial germ cells migrate from the proximal epiblast (yolk sac) through the hindgut and mesentery to the genital ridge and become gonocytes. The precise molecular events underlying transformation to ITGCN are not well understood. The most consistent genetic finding in germ cell tumors is a gain of material from chromosome 12p. The majority of NSGCT and seminomas contain i(12p), an isochromosome composed of two fused short arms of chromosome 12. The remaining i(12p)-negative germ cell tumors also have a gain of 12p sequences in the form of tandem duplications, which may be transposed elsewhere in the genome.

Gain of 12p sequences has been found in ITGCN, indicating that it is an early event in testicular cancer pathogenesis. The acquisition of i(12p) is not thought to be the initiating event, however, because it is preceded by polyploidization.²⁶ Overexpressed genes on 12p are likely to be important, and there are candidate genes on 12p, including several that confer growth advantage (*KRAS2*, *CCND2* [cyclin D2]) and others that establish or maintain the stem cell phenotype (*NANOG*, *DPPA3*, *GDF3*). The exact genes that are critical to this step have not yet been identified.

Seminomas are usually hypotriploid, whereas NSGCT is more commonly hypotriploid.²⁷ Other chromosome regions were found to have nonrandom gains or losses in germ cell tumors with less frequency than 12p. Single-gene mutations are uncommon in germ cell tumors. The *KIT*/kit ligand (*KITLG*) pathway has special relevance for gonadal development. The biologic function of this pathway is broad and includes development of hematopoietic cells, melanocytes, and germ cells.²⁸ *KITLG* is essential for primordial germ cell survival and motility, as are the chemokine *SDF-1* (*CXCL12*) and its receptor *CXCR4*.¹¹ Immunohistochemical markers found on primordial germ cells and gonocytes (*PLAP*, *CD117* [*KIT*], *OCT3/4* [*POU5F1*]) are also found on ITGCN, suggesting a transformation from these cells during fetal development (see Fig. 47.3). The biallelic expression of imprinted genes in germ cell tumors has been reported, showing that they likely arose from primordial germ cells, where the genomic imprinting is temporarily erased.²⁹

The Integrated Molecular Characterization of Testicular Germ Cell Tumors effort published in 2018 analyzed 137 primary testicular germ cell tumors and found a relative paucity of somatic mutations with only *KIT* (18%), *KRAS* (14%), and *NRAS* (4%) having a significant presence with all being present in seminoma.³⁰ The somatic alterations in *KIT* found in germ cell tumors are predicted to upregulate pathway activity. *KITLG* plays a role in determining skin pigmentation and has undergone strong selection in European and Asian populations. Difference in the frequency of risk alleles for *KITLG* between European and African populations may provide an explanation for the difference in germ cell tumor incidence between white Americans and African Americans.

There is evidence that epigenetic regulation of gene expression plays a role in the pathogenesis of germ cell tumors. The DNA methylation patterns are different among histologic types. Global hypomethylation is more common in seminomas than in NSGCT.²⁹ In a study of 16 germ cell tumors, the methylation of CpG islands in NSGCT was similar to that observed in other tumor types, whereas it was virtually absent in seminomas.³¹ Aberrant promoter methylation is generally associated with absent or downregulated expression of the methylated genes. This can result, for example, in the silencing of tumor suppressor genes.²⁹ Methylation has also been correlated with germ cell tumor differentiation. The more differentiated tumors (yolk sac tumor, choriocarcinoma, and teratoma) were consistently hypermethylated, whereas seminoma and GCNIS were hypomethylated.³² Some of the observed methylation patterns may reflect normal development rather than germ cell tumor pathogenesis.

Pluripotency of Embryonal-Like Differentiation in Germ Cell Tumors

Embryonal carcinoma has a six-gene signature (*DNMT3B*, *DPPA4*, *GAL*, *GPC4*, *POU5F1*, *TERF1*), which was detected in three of five independent studies.¹¹ All six of these genes are involved in establishing and maintaining pluripotency. *SOX2* encodes a transcription factor essential for maintaining self-renewal of embryonic stem cells and is upregulated in embryonal carcinoma. Two additional genes that encode transcription factors associated with stem cell pluripotency, *NANOG* and *POU5F1*, are upregulated in both seminoma and embryonal carcinoma.²² Lineage differentiation takes place in NSGCT, mimicking the development of a normal zygote. Thus, embryonal carcinoma can be thought of as the transformed counterpart of embryonic stem cells, displaying self-renewal, pluripotency, and lineage differentiation.

Familial Predisposition to Germ Cell Tumors

Approximately 1.4% of newly diagnosed patients with germ cell tumor report a family history of germ cell tumor.³¹ First-degree male relatives of patients with testicular cancer have a 6- to 10-fold increased risk of developing testicular cancer. These observations point to the likely existence of a hereditary germ cell tumor subset. The inheritable effect is mild, and the most common number of affected relatives in a family is two. The age at diagnosis among familial cases is 2 years to 3 years younger than sporadic cases, and there is a higher incidence of bilateral tumors.

The *gr/gr* deletion on chromosome Y, common among patients with infertility and associated with a two- to threefold increased risk of testicular cancer, was studied as a candidate region for hereditary risk.⁴ The frequency of *gr/gr* variant is low, however, such that it could only account for a small component of hereditary risk. Genome-wide association studies (GWAS) for testicular cancer have identified a total of 44 loci of interest, including signal nucleotide polymorphisms involved in *KIT*-*KITLG* signaling, regulation of DNA repair, apoptosis, sex determination, male germ cell development, and centrosome cycle.^{33,34} Candidate genes repeatedly implicated include *KITLG*, *BAK1*, *SPYR4*, and *DMRT1*. The strongest associated single nucleotide polymorphisms (SNPs) were at 12q22 within the *KITLG* gene, for which there was a greater than 2.5-fold increased risk of disease per major allele. A recently completed meta-analysis of five completed GWAS found 39 SNPs, including 12 new potential susceptibility loci meeting the prespecified level of significance and 13 previously described loci that no longer maintained significance.³⁵ These findings help define the mutational landscape in up to 37% of heritable testicular cancer cases, leaving 63% still unexplained. The strongest associations continue to implicate the aforementioned genes, and eight new candidate loci have been found, including one involving the X chromosome located at Xq28.

Cisplatin Sensitivity and Acquired Resistance in Adult Germ Cell Tumors

Exquisite sensitivity to chemotherapy distinguishes germ cell tumors from most other cancers. Levels of p53 protein are elevated in all germ cell tumors except teratoma.²⁹ To maintain genomic integrity, embryonic stem cells have a high sensitivity to DNA damage, suggesting that the high levels of wild-type p53 seen in germ cell tumors may be intrinsic to their germ cell nature. *TP53* gene mutations are uncommon in germ cell tumors, however, using whole-exome sequencing; mutations in *TP53* or *MDM2* were found in 24% of cisplatin-resistant tumors as compared to 2% of cisplatin-sensitive tumors in a series of 180 patients from a single center, suggesting a role in treatment failure.³⁶

Mutations of *BRAF* including V600E mutations have been detected in a small percentage of NSGCTs. *BRAF* mutations were most prevalent in mediastinal primary tumors, late relapse, and cisplatin-resistant NSGCTs, suggesting a role in chemoresistance. Further study is needed to define the prognostic significance of *BRAF* mutation and whether the V600E mutation can be therapeutically targeted. A high ratio of proapoptotic Bax to antiapoptotic Bcl-2 was found in invasive germ cell tumors and may explain the rapid apoptotic response to DNA-damaging drugs; however, the Bax–Bcl-2 ratio was not associated with treatment outcome.²⁹ Upregulation of the phosphatidylinositol 3-kinase (PI3K)/protein kinase B (AKT) pathway through the platelet-derived growth factor (PDGF)- α and PDGF- β pathways has also been implicated in cisplatin-resistant cells with restoration of sensitivity with downregulation of PDGF signaling.³⁷ Using a hotspot analysis for genetic mutations in DNA extracted from patients with cisplatin-resistant and cisplatin-sensitive germ cell tumors, mutations in *KRAS*, *HRAS*, *PI3K*, and *AKT* were found exclusively in patients with resistance, accounting for approximately 25% of the cases evaluated.³⁸ Finally, the emergence of cisplatin resistance may be intimately associated with the expression of genes and pathways responsible for lineage differentiation in NSGCT, in other words, intrinsic to the germ cell biology rather than the result of specific mutations.

IMMUNOHISTOCHEMICAL MARKERS

Spalt-like transcription factor 4 is expressed in almost all germ cell tumors and has been reported to be positive in GCNIS, classic seminoma, spermatocytic tumor, embryonal carcinoma, yolk sac tumor, choriocarcinoma, and teratoma.^{39,40} Octamer-binding transcription factor 3/4 is variably expressed in GCNIS, classic seminoma, embryonal carcinoma, and yolk sac tumor. Spermatocytic tumor, choriocarcinoma, and teratoma are usually negative for OCT3/4. Cluster of differentiation 117 (KIT) helps highlight GCNIS and classic seminoma. CD30, SRY-box transcription factor 2, and keratin are helpful in the diagnosis of embryonal carcinoma, and SALL4 and glypican-3 are often positive in yolk sac tumor.³⁵ In tumors of unknown primary or those presenting as a retroperitoneal or mediastinal mass, SALL4, OCT3/4, CD117, SOX2, CD30, and low-molecular-weight keratins all may be useful in distinguishing germ cell tumors from non-germ cell tumors.

STAGING

The most widely used system for staging testicular cancer is the tumor, node, metastasis (TNM) classification endorsed by the American Joint Committee on Cancer (AJCC) and the Union for International Cancer Control (UICC) (Tables 47.2 and 47.3).⁴¹ An important distinction for germ cell tumors is the inclusion of “S” classification (S0 to S3), signifying serum tumor marker elevation. There are three stage groupings of TNM/S classifications whereby, in general, stage I disease is confined to the testis, stage II is confined to the retroperitoneal lymph nodes with serum tumor markers in the good-prognosis range (S0–S1), and stage III includes metastases that extend beyond the retroperitoneum or are extranodal in location. Stage III NSGCT also includes any patient with serum tumor markers in the intermediate- or poor-prognosis range (S2–S3).

Patterns of Metastasis

Testicular cancers can undergo both lymphatic and hematogenous dissemination. The lymphatics arising from the testicle accompany the gonadal vessels in the spermatic cord. Some follow the gonadal vessels to their origin, whereas others diverge and drain into the retroperitoneum. The landing zone for metastasis from the right testicle is in the paracaval or interaortocaval lymph nodes just inferior to the renal vessels. The landing zone from the

TABLE 47.2

Definition of Tumor, Node, Metastases (TNM)

TNM Category	Description
PRIMARY TUMOR (T)	
pTX	Primary tumor cannot be assessed
pT0	No evidence of primary tumor (e.g., histologic scar in testis)
pTis	Intratubular germ cell neoplasia (carcinoma in situ)
pT1	Tumor limited to the testis and epididymis without vascular or lymphatic invasion. Tumor may invade into the tunica albuginea but not the tunica vaginalis
pT2	Tumor limited to the testis and epididymis with vascular or lymphatic invasion or tumor extending through the tunica albuginea with involvement of the tunica vaginalis
pT3	Tumor invades the spermatic cord with or without vascular or lymphatic invasion
pT4	Tumor invades the scrotum with or without vascular or lymphatic invasion
Note: Except for pTis and pT4, extent of primary tumor is classified by radical orchiectomy. TX may be used for other categories in the absence of radical orchiectomy.	
REGIONAL LYMPH NODES (N)	
Clinical	
NX	Regional lymph nodes cannot be assessed
N0	No regional lymph node metastasis
N1	Metastasis with a lymph node mass ≥ 2 cm in greatest dimension or multiple lymph nodes, none > 2 cm in greatest dimension
N2	Metastasis with a lymph node mass > 2 cm but not > 5 cm in greatest dimension or multiple lymph nodes, with any one mass > 2 cm but not > 5 cm in greatest dimension
N3	Metastasis with a lymph node mass > 5 cm in greatest dimension
Pathologic (pN)	
pNX	Regional lymph nodes cannot be assessed
pN0	No regional lymph node metastasis
pN1	Metastasis with a lymph node mass ≥ 2 cm in greatest dimension and ≤ 5 nodes positive, with none > 2 cm in greatest dimension
pN2	Metastasis with a lymph node mass > 2 cm but not > 5 cm in greatest dimension; or > 5 nodes positive, with none > 5 cm; or evidence of extranodal extension of tumor
pN3	Metastasis with a lymph node mass > 5 cm in greatest dimension
DISTANT METASTASES (M)	
M0	No distant metastasis
M1	Distant metastasis
M1a	Nonregional nodal or pulmonary metastases
M1b	Distant metastasis other than to nonregional lymph nodes and lungs

TABLE 47.2

Definition of Tumor, Node, Metastases (TNM) (continued)

TNM Category	Description
SERUM TUMOR MARKERS (S)	
SX	Marker studies not available or not performed
S0	Marker study levels within normal limits
S1	LDH <1.5 × N ^a AND hCG (mIU/mL) <5,000 AND AFP (ng/mL) <1,000
S2	LDH 1.5–10 × N OR hCG (mIU/mL) 5,000–50,000 OR AFP (ng/mL) 1,000–10,000
S3	LDH >10 × N OR hCG (mIU/mL) >50,000 OR AFP (ng/mL) >10,000

^aN indicates upper limit of normal for the lactate dehydrogenase (LDH) assay. AFP, α-fetoprotein; hCG, human chorionic gonadotropin; S, serum tumor markers. Used with permission of the American College of Surgeons, Chicago, Illinois. The original source for this information is the AJCC Cancer Staging System (2020).

TABLE 47.3

Stage Grouping

Group	T	N	M	S ^a
Stage 0	pTis	N0	M0	S0
Stage I	pT1–pT4	N0	M0	SX
IA	pT1	N0	M0	S0
IB	pT2	N0	M0	S0
	pT3	N0	M0	S0
	pT4	N0	M0	S0
IS	Any pT/Tx	N0	M0	S1–S3
Stage II	Any pT/Tx	N0–N3	M0	SX
IIA	Any pT/Tx	N1	M0	S0
	Any pT/Tx	N0	M0	S1
IIB	Any pT/Tx	N2	M0	S0
	Any pT/Tx	N2	M0	S1
IIC	Any pT/Tx	N3	M0	S0
	Any pT/Tx	N3	M0	S1
Stage III	Any pT/Tx	Any N	M1	SX
IIIA	Any pT/Tx	Any N	M1a	S0
	Any pT/Tx	Any N	M1a	S1
IIIB	Any pT/Tx	N1–N3	M0	S2
	Any pT/Tx	Any N	M1a	S2
IIIC	Any pT/Tx	N1–N3	M0	S3
	Any pT/Tx	Any N	M1a	S3
	Any pT/Tx	Any N	M1b	Any S

^aMeasured after orchiectomy. Used with permission of the American College of Surgeons, Chicago, Illinois. The original source for this information is the AJCC Cancer Staging System (2020).

left testicle is in the para-aortic lymph nodes just inferior to the left renal vessels. Large-volume disease tends to progress in retrograde fashion to the aortic bifurcation and below, along the iliac vessels.

Seminoma

Seminoma can spread extensively through the lymphatic system to include retroperitoneal, retrocrural, mediastinal, supraclavicular, and cervical lymph nodes, often in the absence of hematogenous metastasis.⁴² Metastasis to lungs (stage IIIA) is common, and metastasis to nonpulmonary organs (stage IIIB) is less common.⁴³ Serum tumor markers do not affect the stage (except in stage IS) or prognosis in seminoma. Hematogenous metastasis to extrapulmonary organs (e.g., bone) in seminoma carries an intermediate prognosis. There is no stage IIIC or poor-prognosis designation in seminoma.⁴⁴

Nonseminomatous Germ Cell Tumors

Similar to seminoma, lymphatic spread is the most common and usually the earliest type of dissemination in NSGCT. Stage groupings depend on both the anatomic extent of disease and serum tumor markers. Stage IIIB is distinguished from Stage II or IIIA on the basis of tumor markers being in the intermediate-prognosis range (S2). Stage IIIC NSGCT carries a poor prognosis, and more often than seminoma, it involves multiple organs such as liver and brain.⁴⁴

Embryonal carcinoma, in some cases, exhibits hematogenous metastasis to lungs or nonpulmonary viscera without clinical involvement of retroperitoneal lymph nodes.⁴⁵ Computed tomography (CT) of the chest is necessary for complete staging workup of tumors that have a high percentage of embryonal carcinoma.

Serum Tumor Markers

Serum tumor markers are an important part of the staging system for germ cell tumors. Three markers, namely AFP, hCG, and total LDH, are considered for establishing the correct prognostic classification (good, intermediate, or poor prognosis).⁴⁴ Markers should be assessed after orchiectomy and before the start of chemotherapy. Markers that are elevated before orchiectomy and then normalize appropriately have no prognostic significance. Markers that do not normalize in a patient without any other clinical evidence of metastatic disease are considered stage IS.⁴¹ In the absence of residual disease, the expected half-life of postoperative serum tumor marker decline is 2 days to 3 days for hCG and 5 days to 7 days for AFP.

Elevated AFP has special significance in seminoma because it is only produced by NSGCT. Germ cell tumors that histologically appear to be pure seminoma with elevated serum AFP are given the clinical diagnosis of NSGCT. hCG can be elevated in either seminoma or NSGCT but has prognostic significance only for NSGCT. Total LDH concentration prior to chemotherapy functions as a prognostic factor in NSGCT and helps to determine stage. In seminoma with bulky metastases, LDH can be markedly elevated but has no prognostic significance.

There are several potential causes of spurious elevation of tumor markers. AFP is not cancer specific; it may be elevated in the presence of liver disease or as a nonspecific chronic elevation. Stability of AFP over time suggests a benign cause. hCG is cancer specific in men, but it is not specific to germ cell tumors. It can be associated with other neoplasms and with exposure to cannabis products. Patients should be questioned about the use of marijuana. A positive hCG test result can also occur as a laboratory artifact in patients with low serum testosterone because of the increased secretion of luteinizing hormone and its sequence similarity to hCG.^{46–48}

Use of microRNA as a Biomarker

MicroRNA (miRNA) are short segments of noncoding RNA that can bind to mRNA posttranslationally and influence the expression of proteins.⁴⁹ Importantly, miRNA is stable in serum and has been the subject of intense exploration in germ cell tumors. Serum

levels of miRNA 371a-3p was found in a multicenter, prospective study of 616 patients with testicular GCT and 258 control patients to have a sensitivity of 90.1%, a specificity of 94%, and a positive predictive value of 97.2%, significantly outperforming the currently used serum tumor markers.⁵⁰ The limitation in miR 371a-3p is the inability to detect teratoma. A second miRNA, miR-375, has been found to be highly expressed in teratoma and yolk sac tumors.³⁰ Other markers with high sensitivities in germ cell tumors include miR-302 and miR-367.⁴⁹ Ongoing studies are exploring the ability to use miR in diagnosis and surveillance and to help with potential surgical decisions.

Clinical Staging

In 2019, the American Urological Association (AUA) provided updated guidelines for the diagnosis and treatment of patients with early-stage testicular cancer through a systemic literature search.⁵¹ Recommendations on staging, workup, and treatment are provided. The guideline does not recommend the use of MRI of the testicle for initial evaluation of testicular cancer. Recommendations are provided regarding testes sparing surgery and counseling patients on the risk of recurrence (high risk) with the potential benefit of fertility preservation. Importantly, the guidelines continue to recommend active surveillance as the preferred management option for stage I seminoma after orchiectomy over chemotherapy or radiotherapy.

The most commonly used methods for detecting metastatic disease are serum tumor markers and CT scan. Positron emission tomography (PET) can be helpful in the staging evaluation of patients with seminoma by distinguishing hypermetabolic lymph node metastases from reactive lymph nodes. PET is not useful in NSGCT, in which CT with oral and intravenous contrast is the preferred technique for detecting retroperitoneal adenopathy.

Pathologic Staging

The T classification is determined by pathology of the orchiectomy specimen.⁴¹ The presence of lymphovascular invasion (LVI) or invasion through the tunica albuginea with involvement of the tunica vaginalis are pT2, invasion of the spermatic cord is pT3, and involvement of the scrotum constitutes pT4 (Table 47.2).

Prophylactic retroperitoneal lymph node dissection (RPLND) is performed for surgical staging in some patients with clinical stage I NSGCT. In such cases, if no metastatic germ cell tumor is found, the stage is pathologic stage I, and when disease is found, it is designated pathologic stage II.

Factors Affecting Outcome

In clinical stage I NSGCT, the presence of LVI (pT2) is associated with approximately 40% to 50% risk of recurrence with surveillance alone. A high percentage of embryonal carcinoma has also been associated with high risk in some series, but embryonal carcinoma is often seen together with LVI and has not been validated as an independent risk factor.

In clinical stage I seminoma, tumor size of 4 cm or larger is associated with approximately 30% risk of recurrence with surveillance alone. Involvement of rete testis has not been validated as a risk factor, although it is often mentioned.⁵²

The classification of patients with metastatic germ cell tumors as good, intermediate, or poor prognosis, based on serum tumor markers and extent of disease, was proposed in 1997 by the International Germ Cell Cancer Collaborative Group (IGCCCG) (Table 47.4).⁴⁴ In this system, the presence or absence of nonpulmonary, extranodal metastasis was validated as an independent prognostic factor for progression-free survival (PFS). For NSGCT, the degree of marker elevation and mediastinal primary (versus testis or retro-

TABLE 47.4

Germ Cell Tumor Risk Classification

Risk Group	Seminoma	Nonseminoma
Good	Any hCG Any LDH Nonpulmonary visceral metastases absent Any primary site	AFP <1,000 ng/mL hCG <5,000 mIU/mL LDH <1.5 × ULN Nonpulmonary visceral metastases absent Gonadal or retroperitoneal primary tumor
Intermediate	Nonpulmonary visceral metastases present Any hCG Any LDH Any primary site	AFP 1,000–10,000 ng/mL hCG 5,000–50,000 mIU/mL LDH 1.5–10.0 × ULN Nonpulmonary visceral metastases absent Gonadal or retroperitoneal primary site
Poor	Does not exist	Mediastinal primary site Nonpulmonary visceral metastases present (e.g., bone, liver, brain) AFP 10,000 ng/mL hCG 50,000 mIU/mL LDH 10 × ULN

AFP, α -fetoprotein; hCG, human chorionic gonadotropin; LDH, lactate dehydrogenase; ULN, upper limit of normal range.

Reprinted with permission from International Germ Cell Consensus Classification: a prognostic factor-based staging system for metastatic germ cell cancers. International Germ Cell Cancer Collaborative Group. *J Clin Oncol* 1997;15(2):594–603.

peritoneal primary) were also validated as prognostic factors. The threshold values for tumor markers (hCG, AFP, and LDH) have been incorporated into the AJCC/UICC staging system as the “S” classification. These prognostic groupings are used to make treatment decisions and are discussed in the following sections.

In the setting of patients with relapsed or refractory disease treated with high-dose chemotherapy (HDCT) and autologous stem cell rescue (ASCT), two prognostic classifications have been proposed. The first, introduced in 1996, has been termed the *Beyer score*. The Beyer score risk stratifies patients into low, intermediate, and high risk subgroups based on the presence of elevated β -hCG greater than 1,000 U/L, primary mediastinal NSGCT, platinum-refractory or absolutely refractory disease, and progressive disease immediately preceding receipt of HDCT.⁵³ A more contemporary scoring system from the International Prognostic Factors Study (IPFS) group found primary site (mediastinal NSGCT again with worst prognosis), prior response to treatment, the progression-free interval from initial cisplatin-based chemotherapy, AFP level, β -hCG level, histology differentiating seminoma from NSGCT, and sites of disease, including lung, bone, and brain metastasis, to stratify patients into very low-, low-, intermediate-, high-, and very high-risk groups.⁵³ The IPFS scoring system is being prospectively analyzed in the ongoing TIGER (Trial of Initial Salvage Chemotherapy for Patients with Germ Cell Tumors) study, which is discussed in a later section.

MANAGEMENT OF CLINICAL STAGE I DISEASE

Virtually all patients with clinical stage I germ cell tumors survive (cancer-specific survival, 98%–99%).⁵² In general, patients managed with observation have the same life expectancy as those who

receive adjuvant intervention. Treatment decisions must therefore be based on considerations of cost, burden of therapy, and patient preference.

Seminoma

Clinical stage I seminoma is grouped into high risk and average risk based on tumor size, but similar options exist for all patients with stage I seminoma, regardless of risk.⁵² Stage I is a common presentation, accounting for approximately 70% to 80% of patients diagnosed with pure seminoma.

Surveillance

The average risk of recurrence with surveillance for stage I seminoma is 15% to 20%. Recurrences are temporally distributed over a 10-year period.⁵⁴ High-risk tumors have a recurrence probability of up to 30%, whereas small tumors may have a risk as low as 12%.⁵⁵ The classic risk factors for recurrence include tumor size larger than 4 cm and rete testis involvement. A recent series found using tumor size as a continuous variable compared with a categorical variable improved performance for assessing risk of recurrence.⁴⁷ Surveillance is heavily dependent on CT scanning because the region at highest risk is in the retroperitoneum, and most seminomas do not secrete tumor markers. At present, both the national consensus guidelines and the 2019 AUA guidelines recommend surveillance as the preferred option for patients with stage I seminoma.⁵¹

Adjuvant Chemotherapy

Carboplatin is a simple and apparently safe form of adjuvant chemotherapy that is very similar in efficacy to prophylactic radiotherapy. In a randomized trial of carboplatin given as a single infusion (area under curve,⁷) versus radiotherapy to the para-aortic lymph nodes, there was no significant difference in PFS (94.7% and 96%, respectively).^{48,56} There was only 1 death from seminoma ($n = 1,447$) reported at a median follow-up period of 6.5 years. In a multicenter retrospective review of patients with stage I seminoma ($n = 185$) treated with adjuvant carboplatin who subsequently relapsed, 85% relapsed within the retroperitoneum and 98% were in the IGCCCG good-prognosis risk group. The majority of patients were cured with standard front-line chemotherapy, and of the 15% of patients who had additional relapse, 3 patients died from the disease.⁵⁷ Recurrences after adjuvant carboplatin were frequently retroperitoneal in location, meaning that follow-up CT scans are mandatory. There has been no reported evidence of second malignant neoplasms (SMNs), and there have been fewer contralateral germ cell tumors reported in the patients treated with chemotherapy versus radiotherapy at medium-term follow-up. Two courses of carboplatin have also been shown to be effective in the adjuvant setting, and in consensus guidelines, one or two cycles of carboplatin are listed as an option for management.

Prophylactic Radiotherapy

Treatment of para-aortic lymph nodes to a dose of 20 Gy was associated with excellent local control approaching 100%. A randomized trial of 20 Gy versus 30 Gy showed no difference in rate of recurrence.⁵⁸ Omission of ipsilateral iliac lymph nodes from the treatment field resulted in less toxicity (infertility, gastrointestinal effects) and minimal loss of efficacy. The recurrence rate after prophylactic radiotherapy for clinical stage I seminoma is approximately 4%, and most of those patients survive with additional treatment (combination chemotherapy). Recurrences tend to occur below the radiation field (pelvic lymph nodes) or above it (lungs). Radiotherapy is contraindicated for patients with horseshoe kidney or inflammatory bowel disease.

Radiotherapy was once popularized because it reduced the number of CT scans that were necessary for follow-up, with a net

reduction in the cost of treatment.⁵⁹ There is, however, a different type of cost with radiotherapy and that is the risk of SMNs.⁶⁰ Studies of testicular cancer survivors 20 or more years after treatment have revealed an increase in midline cancers such as gastrointestinal and genitourinary malignancies.⁶¹ This has brought about a reassessment of prophylactic radiotherapy, specifically, whether it is warranted when 80% of patients with seminoma will be treated unnecessarily, there is a risk of SMN, and there is no survival benefit. Although it continues to be offered, the use of prophylactic radiotherapy for seminoma is declining.⁶²

Nonseminomatous Germ Cell Tumor

Clinical stage I NSGCT is broadly categorized as high risk and average risk. Although there are many differences in presentation, including tumor size, histology, and tumor marker pattern, only LVI has been validated as a risk factor. NSGCT with LVI has a recurrence rate of approximately 50% with observation.⁶³ Most recurrences are seen within 2 to 3 years of the orchiectomy. The average risk is 30%, and most patients with LVI-negative (good-risk) stage I NSGCT probably have a risk of recurrence of less than 30%. The risk of recurrence for LVI-negative stage I NSGCT with predominantly embryonal carcinoma histology, however, may be higher (30%–50%).

Surveillance

Approximately two-thirds of relapses among patients on surveillance are in retroperitoneal lymph nodes, and one-third occur in lungs or by marker elevation alone. Recurrences in nonpulmonary viscera are rare. Most relapses occur within 2 to 3 years of orchiectomy, and patients need to remain on follow-up for at least 5 years. The ability to cure systemic disease with cisplatin-based chemotherapy in those who relapse makes observation an attractive option. It avoids unnecessary therapy for approximately two-thirds of patients.

Criteria for selecting patients for observation have been suggested in the literature and consensus guidelines. Selection based on whether the patient is “reliable” (i.e., likely to be compliant with follow-up) was once the prevalent view, but this has been largely replaced by a more objective risk-adapted approach. Observation is the standard of care for patients with stage IA NSGCT. The alternative is nerve-sparing RPLND, which is an unnecessary procedure for the majority of these patients. There is no subgroup of stage IA for whom RPLND is preferred, but patients who choose observation should agree to be compliant with follow-up and to receive chemotherapy in the event of recurrence. Patients with stage IA embryonal carcinoma-predominant tumors are probably the least likely to benefit from RPLND because of the tendency for hematogenous spread and associated risk of recurrence in lungs.⁶⁴

Most patients with LVI-negative, clinical stage I NSGCT are candidates for observation. There is less of a consensus on management of patients with pT2 to pT4 tumors (including LVI-positive tumors). A minority of patients with pT2 tumors may choose observation, but they must understand that the risk of recurrence is 50%, and they should agree (as in stage IA) to comply with follow-up. The preferred treatment for LVI-positive or advanced T classification NSGCT is adjuvant chemotherapy.

Adjuvant Chemotherapy

Primary chemotherapy for NSGCT consists of bleomycin, etoposide, and cisplatin (BEP) (Table 47.5).^{65,66} There are several published studies using either one or two courses of adjuvant BEP or similar regimens in patients with clinical stage I NSGCT.^{67–71} A phase II study by the Medical Research Council found 98% relapse-free survival after two courses of bleomycin, vincristine, and cisplatin, although chemotherapy-induced neurotoxicity (CIN) remained a problem.⁷⁰ A recently completed phase 3 study of one cycle of adjuvant BEP with high risk features found the

TABLE 47.5**Chemotherapy Regimens for Stage II or III Germ Cell Tumors****Previously Untreated: Good Risk^a**

EP (etoposide, cisplatin): four cycles administered at 21-d intervals
 Etoposide 100 mg/m² IV daily × 5 d, cisplatin 20 mg/m² IV daily × 5 d

BEP (bleomycin, etoposide, cisplatin): three cycles administered at 21-d intervals
 Bleomycin 30 units IV weekly (days 1,8,15), etoposide 100 mg/m² IV daily × 5 d, cisplatin 20 mg/m² IV daily × 5 d

Previously Untreated: Intermediate and Poor Risk

BEP: four cycles administered at 21-d intervals
 Bleomycin 30 units IV weekly (days 1,8,15), etoposide 100 mg/m² IV daily × 5 d, cisplatin 20 mg/m² IV daily × 5 d

VIP (etoposide, ifosfamide, cisplatin): four cycles administered at 21-d intervals
 Etoposide 100 mg/m² IV daily × 5 d, ifosfamide 1200 mg/m² IV daily × 5 d, cisplatin 20 mg/m² IV daily × 5 d, Mesna 400 mg/m² IV every 8 h × 5 d

Previously Treated: First-line Salvage Therapy

VeIP (vinblastine, ifosfamide, cisplatin): four cycles administered at 21-d intervals
 Vinblastine 0.11 mg/kg IV days 1 and 2, ifosfamide 1.2 g/m² IV daily × 5 d, cisplatin 20 mg/m² IV daily × 5 d, MESNA 400 mg/m² IV every 8 h × 5 d

TIP (paclitaxel, ifosfamide, cisplatin): four cycles administered at 21-d intervals (preferred)
 Paclitaxel 250 mg/m² IV by continuous infusion over 24 h on day 1, ifosfamide 1500 mg/m² IV daily on days 2–5, cisplatin 20 mg/m² IV daily × 5 d, Mesna 400 mg/m² IV every 8 h × 5 d

^aGood-risk regimens (bleomycin, etoposide, and cisplatin or etoposide and cisplatin) can also be used for stage I adjuvant treatment or stage IS. IV, intravenous.

2 year relapse rate to be 1.3%.⁷¹ One limitation of adjuvant chemotherapy is the continuing risk of growing teratoma syndrome. Only RPLND can remove foci of teratoma from the retroperitoneum, where they may persist and grow following chemotherapy.⁷² National consensus guidelines endorses surveillance for stage I without risk factors and either surveillance, one cycle of BEP chemotherapy, or nerve-sparing RPLND for patients with risk factors.

Retroperitoneal Lymph Node Dissection

Retroperitoneal lymph node dissection performed in patients with clinical stage I NSGCT is both diagnostic and therapeutic: it is done with the intent to accurately stage the patient (as pathologic stage I or stage II) and remove all viable disease. RPLND is curative for teratoma, which has been reported in 21% to 30% of cases with viable disease, as well as about 90% of patients with pN1 disease with viable germ cell tumor.⁷³ The mortality rate from RPLND is less than 1%; minor complications include prolonged ileus, wound infection, and lymphocele. Major complications (hemorrhage, ureteral injury, chylous ascites, pulmonary embolus, wound dehiscence, bowel obstruction) are rare. A notable potential long-term morbidity is sympathetic nerve damage leading to retrograde ejaculation.⁷⁴

For optimal outcomes, RPLND should be performed at a referral center with an experienced surgeon. A bilateral infrahilar RPLND includes the precaval, retrocaval, paracaval, interaortocaval, retroaortic, preaortic, para-aortic, and common iliac lymph nodes.

There are two types of nerve-sparing RPLND: the modified template RPLND and nerve-dissection bilateral RPLND.^{73,75,76} The template dissection helps the surgeon avoid regions where the risk of metastasis may be less. The nerve-dissection technique actively identifies and preserves both sympathetic chains, postganglionic sympathetic fibers, and the hypogastric plexus, which are necessary for anterograde ejaculation. The reported incidence of retrograde ejaculation with this technique is less than 5%. Another innovation is the robot-assisted RPLND, in which robotic instruments are inserted through a series of trocar entry sites and controlled remotely by the surgeon.^{77–80} Patients have less pain, shorter hospitalization times, rapid recovery, and smaller surgical scars.

The relapse rate after RPLND is variable. It may be as low as 4% with a low-risk patient and an experienced surgeon. Many series include patients who also received adjuvant chemotherapy, typically with two courses of BEP, for pathologic stage II disease. In these patients, the success of RPLND can be confounded by the administration of chemotherapy in patients whose patients showed viable germ cell tumor (i.e., not teratoma). LVI-positive and other high-risk patients have higher reported failure rates (10%–14%) with primary RPLND.⁷² The greatest advantage of RPLND is the early control of teratoma when it exists. A randomized trial of RPLND versus one course of BEP found a recurrence rate of 8.3% in the RPLND arm and 1.1% in the BEP arm.⁶⁸ With median follow-up period of 4.7 years, only one patient had growing teratoma after BEP. Limitations of the study were that unilateral dissections were performed (nonstandard) and that the experience level of surgeons varied, as it was a community-based study leading to a higher rate of infield relapse compared with high-volume, single-center series. The low incidence (1%–2%) of growing teratoma seen among patients on observation or adjuvant chemotherapy is lower than one would expect based on the results of surgical staging and suggests that not all teratomas have the same potential for growth or malignant transformation. Quality of life between the two arms was compared over 3 years, and by 6 months posttreatment, patients in both arms had similarly recovered to baseline.⁸¹

MANAGEMENT OF CLINICAL STAGE II DISEASE (LOW TUMOR BURDEN)

The anatomic definition of stage II germ cell tumor is metastasis confined to retroperitoneal lymph nodes. For NSGCT, there is the further requirement that tumor markers be in the good-prognosis range (S0–S1).⁴¹ For both seminoma and NSGCT, the majority of these patients are cured with standard treatment.

Seminoma

Seminoma is exquisitely sensitive to both chemotherapy and radiation. Radiotherapy to the retroperitoneum with a boost to the involved nodes is standard for disease with a transverse measurement of 3 cm or less.^{82,83} The fields include the landing zone and proximal ipsilateral iliac lymph nodes (dog-leg field). Radiotherapy is curative in 80% to 90% of patients, with recurrence largely caused by occult disease outside the radiation field.⁸⁴ Nevertheless, prophylactic radiotherapy to the mediastinum or supraclavicular nodes is not recommended because of toxicity concerns.⁸⁵ There is an excellent rate of success with combination chemotherapy in patients who do relapse after radiotherapy.⁸⁶

For patients with bulky (>3 cm transverse dimension) retroperitoneal disease, primary treatment with cisplatin-based chemotherapy is standard, followed by radiotherapy for consolidation to a residual mass greater than 3 cm.⁸⁶ As in the adjuvant setting, patients with horseshoe kidney or inflammatory bowel disease should not receive radiotherapy; chemotherapy is the primary treatment for such patients.

The role of primary RPLND in patients with stage IIA/B seminoma is being evaluated in two ongoing clinical trials. The PRIMETEST trial, in which the interim analysis with a median follow-up period of 24 months showed that 17 of 22 patients (77%) are free of recurrence.⁸⁷ The SEMS (Surgery in Early Metastatic Seminoma) trial recently reported on 55 patients who underwent RPLND (14 were initially stage I and later developed isolated retroperitoneal disease, and 41 were clinical stage IIA-B at presentation). At a median follow-up period of 24 months, there were 10 recurrences. The overall recurrence rate was 18% with a median time to recurrence of 8 months. The 2-year recurrence-free survival rate was 87%, and the overall survival (OS) rate was 100%.⁸⁸ Currently, this approach is still considered experimental and should only be considered in the context of a clinical trial.

Nonseminomatous Germ Cell Tumor (Low Tumor Burden)

Stage IS

Serum tumor markers that do not normalize after radical orchiectomy are evidence of micrometastases. Treatment has classically been with adjuvant chemotherapy with either three cycles of BEP or four cycles of etoposide and cisplatin (EP). A move toward active surveillance for these patients has become more common. A recent retrospective review using the National Cancer Database found an excellent and similar survival rate ($\geq 95\%$) for both patients with stage IS NSGCT and patients with seminoma who either received adjuvant chemotherapy or had been followed by active surveillance.⁸⁹ Given the rarity of this stage, in the modern era of cross-sectional imaging, these data have strengthened the consideration of surveillance for these patients.

Pathologic Stage II After Retroperitoneal Lymph Node Dissection

In the case of low-volume metastatic NSGCT, a prophylactic RPLND may be curative without additional therapy.^{64,90} Patients with fewer than six lymph nodes involved, with no focus greater than 2 cm, and without extranodal extension (pN1) have an estimated 10% to 20% risk of recurrence and can be managed with surveillance. Adjuvant chemotherapy with BEP or EP (two courses) is an option for selected patients based on their access to follow-up, psychological factors, and tumor histology. Although there is not strong evidence to support adjuvant decision making based on the histology, metastases with predominantly teratoma or yolk sac tumor may have lower post-RPLND risk of recurrence than those that have predominantly embryonal carcinoma or choriocarcinoma. Nevertheless, surveillance and treatment with three or four courses of chemotherapy at recurrence are likely to yield a similar OS as adjuvant treatment. Adjuvant chemotherapy should be recommended for patients with pN2 to pN3 disease. Consensus guidelines currently recommend two cycles of chemotherapy for patients with pN2 disease and full treatment for patients with N3 disease.

Clinical Stage IIA

Primary chemotherapy with three courses of BEP is the standard treatment for patients with enlarged retroperitoneal lymph nodes and elevated serum tumor markers (S1).⁶⁵ Patients with enlarged lymph nodes and normal serum tumor markers (N1, S0) may be appropriate for nerve-sparing bilateral RPLND or chemotherapy. A primary RPLND can be considered if the primary tumor contains teratoma and is not embryonal carcinoma predominant, tumor markers are not elevated, and there is no back pain. Elevated tumor markers, back pain, or nodal size larger than 2 cm suggests multifocal or unresectable disease, and chemotherapy is preferred.⁹¹

Embryonal carcinoma can be marker negative and requires chemotherapy. If metastatic embryonal carcinoma is suspected on clinical grounds (enlarged retroperitoneal lymph nodes), then chemotherapy is the treatment of choice, whether or not the markers are elevated.

MANAGEMENT OF STAGE II DISEASE WITH HIGH TUMOR BURDEN AND STAGE III DISEASE

Patients with metastatic germ cell tumor and high tumor burden are curable, but they can be critically ill at presentation. The chance of survival improves with early recognition of the diagnosis by medical providers, often on clinical grounds, and the prompt administration of chemotherapy. Whenever possible, patients with stage IIIC NSGCT should be treated by an experienced team at a tertiary care center.^{86,92}

Good Prognosis

The IGCCCG risk classification identifies a good-prognosis subset with OS rate of 90% to 95% with standard therapy.⁴⁴ These patients include most of those with metastatic seminoma, including mediastinal seminoma, excluding only seminoma with nonpulmonary visceral metastasis, and NSGCT stages II and IIIA (metastasis confined to lymph nodes and lungs and tumor markers below the intermediate-risk [S2] or high-risk [S3] levels).⁹³ For good-prognosis NSGCT, three courses of BEP result in normalization of tumor markers for the majority of patients.⁶⁵ Postchemotherapy RPLND is necessary for patients with NSGCT who have a residual mass 1 cm or larger because it can harbor teratoma or other viable disease.⁹⁴

An alternative to three courses of BEP, for good-prognosis patients only, is EP given for four courses.⁹⁵⁻⁹⁷ For selected patients, the added risk of CIN and other complications from a fourth course of EP may be balanced by avoidance of the pulmonary toxicity risk of bleomycin. In practice, patients with seminoma are less tolerant of bleomycin because of older age, and giving four courses of EP is a reasonable standard for the majority of patients.

Intermediate and Poor Prognosis

The intermediate-prognosis subset (stage IIIB) accounts for 25% of patients with metastatic germ cell tumors and has an OS of approximately 75% with standard therapy.⁴⁴ The poor-prognosis group (stage IIIC) is exclusively NSGCT, accounts for 15% of metastatic germ cell tumors, and has 5-year PFS rate of 45%. This group includes all patients with mediastinal primary NSGCT, nonpulmonary visceral metastasis, or tumor markers in the S3 range. BEP for four courses is the standard primary treatment for patients with stage IIIB and IIIC germ cell tumors.⁹⁸⁻¹⁰⁰ For patients with seminoma with an intermediate prognosis who are unlikely to tolerate bleomycin, the substitution of ifosfamide to etoposide and cisplatin (VIP) is a reasonable standard for these patients.⁸⁶ In NSGCT, VIP can also be used instead of BEP if there is an increased risk of bleomycin lung toxicity because of respiratory distress, age 50 years or older, smoking or other chronic respiratory disease, or anticipated major thoracic surgery. A recently completed phase II study randomized 89 patients with poor risk NSGCT to receive CBOP/BEP versus BEP, and after a median follow-up period of 63 months, the 3-year PFS was 55.7% (95% confidence interval [CI], 39.7%, 69.0%) for CBOP/BEP and 38.7% (95% CI, 24.7%, 52.4%) for BEP (hazard ratio, 0.59 [0.33, 1.06]; $P = 0.079$), warranting consideration of a larger study to evaluate this regimen.¹⁰¹

Personalized Strategy Based on Tumor Marker Decline

Serum tumor markers (AFP, hCG) virtually always decline during the initial two to three cycles of chemotherapy. Failure of either marker to normalize is a well-recognized feature of chemotherapy resistance.¹⁰² The rate of tumor marker decline has also been studied as a predictor of poor outcome. For patients presenting with stage IIIC NSGCT, it is possible to identify a subgroup of approximately 25% of patients who will do comparatively well and a larger group of approximately 75% whose outcome with standard therapy is poor.¹⁰³ This observation led to the Groupe D'étude des Tumeurs Urogénitales (GETUG) 13 phase III clinical trial in which patients with stage IIIC NSGCT received BEP in the first cycle. Based on the tumor marker decline in the first cycle, those with favorable decline remained on BEP (four courses total), and the rest were randomized (1:1) to BEP or an intensified regimen. The patients with unfavorable marker decline had superior 3-year PFS if provided the intensified regimen (59% vs 48%).¹⁰⁴

Central Nervous System Metastases

Patients with brain metastasis are curable.¹⁰⁵ Imaging of the brain, preferably with MRI, is appropriate at baseline for patients with stage II or III disease and should be considered mandatory for patients with a significantly elevated β -hCG given the concern for choriocarcinoma syndrome. The initial management for most patients with asymptomatic brain metastases is with systemic chemotherapy. Responses tend to occur swiftly (Fig. 47.4), but there is a risk of intracranial hemorrhage. The risk of bleeding can be minimized by modifying the first cycle of chemotherapy (e.g., EP given for 3 days). Tumors with active bleeding may require craniotomy. Radiotherapy is useful for postchemotherapy consolidation of residual lesions in the brain. Gamma knife is preferable to whole-brain radiotherapy in these patients with potentially long survival times and risk of cognitive impairment.

The brain is a potential sanctuary site. This can manifest as a solitary recurrence in the brain shortly after the completion of chemotherapy, for which surgical resection may be curative. In the

GETUG 13 study, the brain served as a common site of first progression, occurring in 21 of 109 patients (19%), with recurrence suggesting patients with poor risk disease in remission should have consideration of brain MRI during surveillance.¹⁰⁶

Choriocarcinoma Syndrome

Metastatic choriocarcinoma is characterized by rapid hematogenous spread.¹⁰⁷ It usually starts as a component of a mixed testicular NSGCT but can then proliferate and dominate the clinical picture. Choriocarcinoma syndrome is recognizable by very high serum hCG levels in the range of 10^5 to 10^6 mIU/mL and occasionally over 1,000,000 mIU/mL, a testicular mass (or mediastinal mass), diffuse lung metastases, involvement of nonpulmonary viscera (brain, liver), tumor hemorrhage (hemoptysis, hemoperitoneum, intracranial bleed), and hyperthyroidism. The hyperthyroidism occurs at very high levels of hCG because hCG has structural similarity to thyroid-stimulating hormone.^{108,109}

Patients with choriocarcinoma syndrome have a high rate of mortality caused by complications such as hemorrhage or as a result of recurrent or refractory disease. At the time of diagnosis, however, the clinical condition rapidly stabilizes with the administration of chemotherapy. It is not recommended to use bleomycin for the first course in most cases because of high-volume pulmonary metastases and the risk of pulmonary hemorrhage. To avoid destabilizing the patient, EP chemotherapy can be shortened to 3 days for the first course only. For subsequent courses, EP is not sufficient for stage IIIC NSGCT, and either BEP or an ifosfamide-containing regimen should be given. Administration of a beta-blocker during the first course alleviates symptoms of hyperthyroidism (hypertension, tachycardia).

A male patient with testicular mass or anterior mediastinal mass, serum hCG greater than 50,000 mIU/mL, and a clinical picture of choriocarcinoma syndrome does not require orchiectomy or biopsy before the start of treatment. The clinical diagnosis of choriocarcinoma should be recognized and treated as a medical emergency (see Fig. 47.4). Immediate chemotherapy offers the best chance for survival. Resection of the involved testis should be performed between cycles when the patient has stabilized or at the time of RPLND.

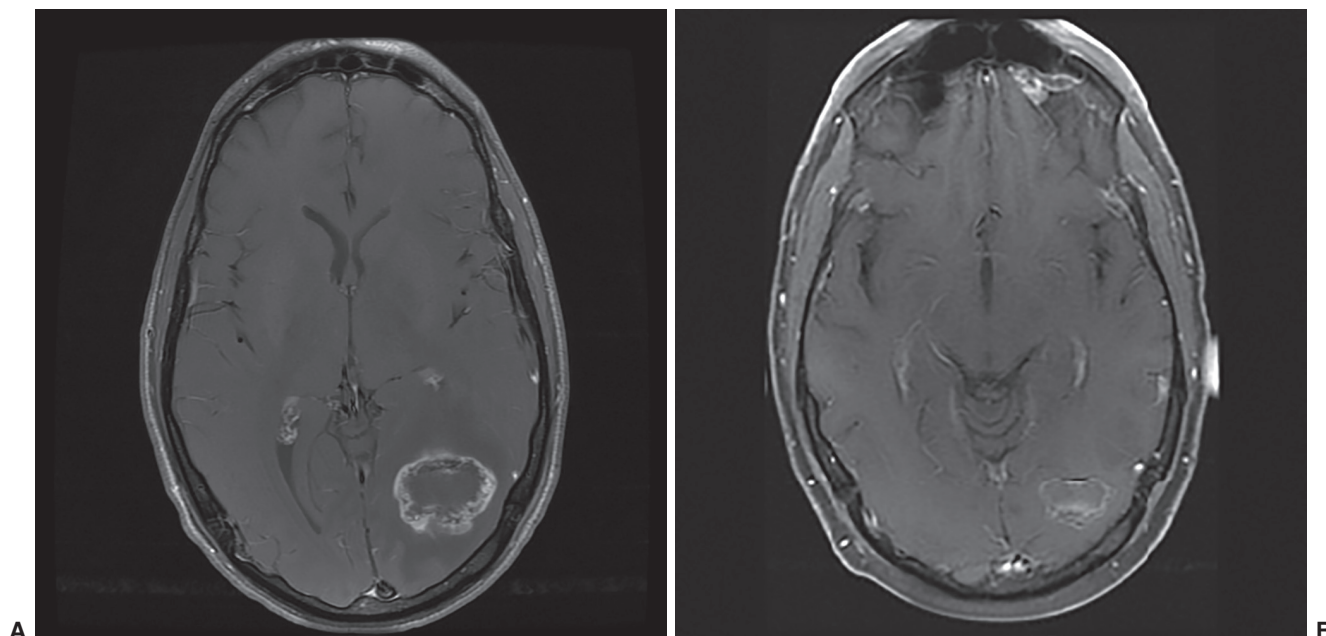


Figure 47.4 Brain metastasis in a patient with testicular nonseminomatous germ cell tumor and clinical features of choriocarcinoma syndrome. Magnetic resonance images were acquired at diagnosis (**A**) and after three cycles (**B**) of cisplatin-based chemotherapy (9 weeks) using modified regimens to avoid thrombocytopenia.

Mediastinal Nonseminomatous Germ Cell Tumor

Extragenital germ cell tumors are the result of arrested migration of germ cells along the urogenital ridge during embryogenesis. This aberrant germinal tissue is usually located along the cranio-caudal axis in adult life, and malignant transformation can occur in both women and men. The most common presentation is in the anterior mediastinum of a man.¹¹⁰ Patients with Klinefelter syndrome are at increased risk for mediastinal germ cell tumors.^{111,112}

Mediastinal NSGCT is curable but at a lower rate than most testicular germ cell tumors.¹¹³ The 3-year PFS rate described in a modern therapy series was 48% to 54%.^{114–117} The first-line chemotherapy is the same as for other stage IIIC germ cell tumors, consisting of BEP for four courses for patients with good pulmonary function and VIP for four courses for those who are unlikely to tolerate bleomycin. Some authors have advocated that VIP is the preferred regimen because most patients undergo postchemotherapy thoracic surgery.¹¹⁸ In a prospective multicenter study, however, 66 patients with mediastinal NSGCT were treated with bleomycin-containing regimens, and excessive pulmonary complications were not seen.¹⁰⁴ Patients with mediastinal primary tumors have a high incidence of viable germ cell malignancy, teratoma, and transformation to somatic malignancy in postchemotherapy resections, so surgical consolidation is essential.^{114,119}

Management of Residual Mass

Patients with high tumor burden frequently have one or more sites of residual disease after chemotherapy and normalization of tumor markers. The management of a residual mass is critical to the curative management of germ cell tumors. There are different treatment considerations based on the setting of seminoma or NSGCT and the size of the lesion.

Residual Nodal Smaller Than 1 cm

For either seminoma or NSGCT, a radiographic complete response does not require consolidative treatment. The size criterion for absence of residual mass in a site of lymph node metastasis is a transverse dimension less than 1 cm (on CT imaging). There is a potential benefit to some patients with NSGCT from the surgical consolidation of residual masses greater than 0.5 cm and less than 1.0 cm, based on the possibility of teratoma.⁷² By some estimates, as many as 20% of nodes smaller than 0.5 cm and 29% of nodes smaller than 1.0 cm harbor teratoma. As noted previously, however, the incidence of growing teratoma syndrome among patients observed without RPLND is lower (1%–2%), and these can be detected on follow-up with CT imaging. Most are treated successfully, and RPLND for a residual mass smaller than 1 cm is the exception rather than the rule.

Residual Mass Larger Than 1 cm

A persistent mass in a nodal site after chemotherapy is of concern for residual germ cell malignancy in both seminoma and NSGCT and for teratoma in NSGCT. The same applies for residual masses in extranodal organs.

For NSGCT, the standard management of a residual mass larger than 1 cm in the retroperitoneum is a bilateral RPLND.⁹⁴ Modified template RPLND is not appropriate in this setting because it may leave behind disease in 7% to 32% of cases. Robot-assisted RPLND is an option for very well-selected patients, resulting in less short-term morbidity of the procedure compared with an open RPLND.^{78,120} The principal long-term morbidity is retrograde ejaculation.¹²¹ After the standard three or four courses of BEP, viable germ cell malignancy is found in only 15% of specimens. Forty percent contain teratoma, and 45% contain only necrosis and fibrosis. There is no reliable method for determining preoperatively whether teratoma is present, though microRNA

does hold potential promise as a future option to select patients for surgery. About 25% to 45% of patients with no teratoma in the primary tumor can still have teratoma in a residual mass. Biopsy is useless for ruling out the presence of teratoma because of sampling error. Complete excision is the standard of care. For the minority of patients with viable germ cell malignancy (other than teratoma) in a residual mass, the standard treatment is postoperative administration of two additional courses of chemotherapy. Residual teratoma requires no further therapy.

Seminoma

Unlike NSGCT, seminoma does not produce teratoma. A residual mass is common because of the fibrotic reaction that occurs in the treated lymph nodes. Surgery is technically difficult and in most cases unnecessary. Patients with very high-volume seminoma or residual mass larger than 3 cm can relapse from residual seminoma. The optimal consolidation for these patients is an unsettled question. Patients with residual mass smaller than 3 cm can remain on observation. For those with a mass larger than 3 cm, radiotherapy is one means to reduce the risk of relapse. An OS benefit from postchemotherapy radiation has not been demonstrated. This may be in part because of the overall high survival rate for seminoma and the relatively small proportion of patients at risk for relapse after chemotherapy. Consensus guidelines recommend surgery (RPLND), if technically feasible, or biopsy with resection if positive for residual masses larger than 3 cm with PET positivity.

MANAGEMENT OF RECURRENT DISEASE

Patients in first relapse after BEP chemotherapy can be successfully salvaged in approximately 50% of cases. The relapse-free survival rate with HDCT and ASCT in one retrospective study was 60% (90 of 149 patients with recurrent or refractory NSGCT). Whether this was an improvement in outcome over conventional chemotherapy or a reflection of patient selection is unknown. A randomized trial (the TIGER study) for patients in first relapse receiving paclitaxel, ifosfamide, and cisplatin (TIP) for four courses versus HDCT-ASCT using the TI-CE regimen (paclitaxel plus ifosfamide followed by high-dose carboplatin plus etoposide with stem cell support) is currently open and enrolling patients with the hope of addressing this remaining question.

Conventional-Dose Chemotherapy

Standard-dose regimens have resulted in a complete response rate of 35% to 70% in the second-line setting without the use of HDCT-ASCT. The most effective regimens for first recurrence after BEP have been combinations with ifosfamide and cisplatin.^{122,123} Examples are vinblastine, ifosfamide, and cisplatin and TIP, typically given for four courses with consolidation surgery (see Table 47.5).^{122,124}

High-Dose Chemotherapy and Stem Cell Transplant

In the most recent series from patients treated at Indiana University from 2004 to 2014, 364 patients received two cycles of high-dose carboplatin plus etoposide (“tandem transplant”).¹²⁵ At a median follow-up period of 3.3 years, the 2-year PFS rate was 60%, and the 2-year OS rate was 66%. This retrospective review excluded mediastinal NSGCT, late relapses, and other nongonadal primary sites (unfavorable). In a prospective trial from the Memorial Sloan Kettering Cancer Center, 81 patients received one to two cycles of paclitaxel plus ifosfamide followed by three cycles of high-dose

carboplatin plus etoposide as first salvage (TI-CE).¹²⁶ At a median period of 5 years, 56% of patients remained free of disease. This study excluded patients with a testicular primary tumor who either had a prior complete response to first-line chemotherapy or a partial response with negative markers but included patients with mediastinal NSGCT, late relapse, and other nongonadal primary sites. A recently updated prospective trial from MD Anderson Cancer Center for patients with platinum-refractory or high-risk criteria per the Beyer score found significant activity of a novel HDCT regimen using gemcitabine, docetaxel, melphalan, and carboplatin in the first HDCT followed by ifosfamide, carboplatin, and etoposide during second HDCT, which will need to be evaluated in future randomized clinical trials.¹²⁷

Second and Subsequent Relapse

Patients with recurrence after conventional second-line chemotherapy such as vinblastine, ifosfamide, and cisplatin or TIP can be considered for HDCT-ASCT.¹²⁸ Compared with patients in first relapse, however, these patients have more chemotherapy sequelae such as CIN, renal insufficiency, and decreased performance status, leading to a high complication rate and mortality rate from HDCT-ASCT. Conventional chemotherapy after second-line treatment or HDCT-ASCT is usually palliative. Regimens endorsed by national consensus guidelines include gemcitabine and oxaliplatin; gemcitabine and paclitaxel; gemcitabine, oxaliplatin, and paclitaxel; and oral etoposide.^{129–131}

Molecularly targeted therapies do not yet have an established role in the treatment of germ cell tumors. An inhibitor of cyclin-dependent kinase 4/6 showed activity in patients with unresectable teratoma, leading to clinical trials.¹³² Patients with chemotherapy-refractory germ cell tumors should be encouraged to participate in clinical trials.

Defining the immune landscape has been of increasing interest in germ cell tumors given the introduction of immune checkpoint therapy to the cancer treatment armamentarium across a broad range of liquid and solid tumors. The initial clinical experience with the anti-programmed cell death protein 1 (PD-1) agent pembrolizumab was disappointing, with none of 10 patients responding in the platinum-refractory setting in one study.^{133,134} Additional studies evaluating the activity of avelumab, nivolumab plus ipilimumab, and durvalumab with or without tremelimumab have also yielded disappointing results.^{135,136}

Surgery After Salvage Chemotherapy

Residual lesions that persist after chemotherapy should be resected whenever feasible. There is often viable tumor in the setting of recurrent or refractory disease, even when tumor markers have normalized on chemotherapy.¹³⁷ When tumor markers decline to a plateau but do not normalize, surgery to remove all viable disease is a consideration. Surgical salvage is successful in approximately 20% of patients.^{138,139} Patients with elevated AFP and normal hCG have better prognosis with surgery than patients with elevated or increasing hCG. Adjuvant chemotherapy is not known to improve the outcome when surgery reveals viable germ cell malignancy after salvage chemotherapy.

Late Recurrence Nonseminomatous Germ Cell Tumor

Most NSGCT recurrences after chemotherapy are seen within 2 to 3 years. Only approximately 2% of patients experience a relapse after 2 years. These late recurrences appear to be less sensitive to subsequent chemotherapy than the earlier recurrences.^{140,141} Surgery can be curative in cases that are slowly growing and anatomically resectable. Treatment must be individualized. Although late

recurrences are less chemosensitive as a group, complete responses have been described. Surgery should always be considered and integrated as a component of the overall treatment plan.

TREATMENT SEQUELAE

Both chemotherapy and radiotherapy have long-term adverse effects on testicular cancer survivors.^{2,60,61,85} With posttreatment life expectancy of 40 years or more, the morbidity attributable to chronic and late effects of treatment can exceed that of the cancer itself.

Cardiovascular Effects

One of the most significant delayed effects of chemotherapy is an increased risk of cardiovascular disease.⁸⁵ The risk of having a cardiac event is two- to sevenfold higher in testicular cancer survivors who received cisplatin-based chemotherapy than the general population.^{142,143} Other risk factors such as hyperlipidemia, obesity, and hypertension (metabolic syndrome) are also more common among testicular cancer survivors. The health and longevity of testicular cancer survivors can be maximized through early treatment of hypertension and hyperlipidemia and by interventions such as diet, exercise, and tobacco cessation. A recent randomized phase II study using a supervised high-intensity aerobic interval training program compared to usual care in germ cell tumor survivors found 12 weeks of this intensive exercise to correlate with improvement across a variety of parameters that have been used previously to correlate positively with cardiac health.¹⁴⁴ Although the long-term benefits of such a program are unknown, reducing the emergence of metabolic syndrome in the patient population appears to be clearly advantageous.

Chemotherapy-Induced Neurotoxicity

Peripheral neurotoxicity is a common adverse effect of cisplatin. Approximately 20% to 40% of patients treated with neurotoxic chemotherapy drugs develop CIN, which can cause painful and permanent sensory disturbance. Drug therapy for patients with CIN is only partially effective in relieving symptoms.¹⁴⁵ The mechanisms of CIN are not well understood and appear to include damage to neuronal cell bodies in dorsal root ganglia.¹⁴⁶ Circulating platinum levels remain elevated for many years after chemotherapy, and it is not known whether this also contributes to long-term morbidity.¹⁴⁷ The severity of CIN is influenced by the cumulative dose of cisplatin, exposure to other neurotoxic drugs (e.g., paclitaxel), and other medical conditions such as diabetes.

Hypogonadism and Infertility

Chemotherapy can damage the germinal epithelium and increase the risk of abnormal sperm morphology, motility, and number. Oligospermia has been associated with prior radiotherapy for seminoma, presumably caused by scatter radiation to the contralateral testis. It is also recognized that at the time of diagnosis, the percentage of men with germ cell tumor who are subfertile or infertile is greater than the general male population. Thus, sperm banking should be offered to all patients undergoing chemotherapy or radiotherapy and to patients undergoing RPLND because they are at risk for retrograde ejaculation.

The risk of infertility from treatment is proportional to the type and duration of treatment. In a study of paternity after treatment, 71% of unselected testicular cancer survivors were successful at 15 years. For treatment subgroups, successful paternity rates were 81% with surveillance, 77% after RPLND, 65% after radiotherapy, 62% after chemotherapy, and 38% after high-dose salvage chemotherapy. The risk to patients receiving fewer than four cycles of

chemotherapy has not been studied, particularly in the adjuvant setting (one or two cycles) or with carboplatin in the setting of seminoma. Although the risk of infertility may be lower in the adjuvant setting, these patients should receive fertility counseling and an opportunity for cryopreservation of semen.

Hypogonadism or low testosterone is also a common finding. Testicular dysfunction is more common among testicular cancer survivors than the general male population. Persistent low testosterone in a patient who has completed therapy is an indication of a functional deficit in the contralateral testis. Testosterone replacement therapy can prevent complications such as weight gain, gynecomastia, erectile dysfunction, loss of libido, fatigue, depression, and osteoporosis.

Ototoxicity

Cisplatin can result in permanent, bilateral sensorineural hearing loss in 19% to 77% of patients and tinnitus in 19% to 42% of patients.¹⁴⁸ The incidence and severity of ototoxicity are related to the cumulative dose and dose intensity of cisplatin.¹⁴⁹ The mechanism is thought to be through overproduction of reactive oxygen species in the cochlea, causing irreversible free radical-related apoptosis of outer hair cells, spiral ganglion cells, and the stria vascularis.¹⁴⁸ There may be genetic underpinnings to the susceptibility to ototoxicity from chemotherapy. A recent series from Canada found the presence of the genetic variant of *ACYP2* was associated with ototoxicity with cisplatin and *WFS1* was associated with hearing loss from any cause.¹⁵⁰ There is no effective method for treating or preventing cisplatin-induced ototoxicity.¹⁵¹

Psychosocial Functioning

There are potential short- and long-term psychological consequences in the posttreatment period.¹⁵² Anxiety and depression are common in the first 6 months, but most patients are well adjusted by 1 year. Certain patients (10%–30%) continue to have moderate to severe nervousness, anxiety, or depression. Patients who have sexual difficulties, are unemployed, or have financial difficulties appear to be at greatest risk. Strain in relationships can be caused by a perception of sexual dysfunction, although some studies suggest that in married couples, the patient is more concerned than the spouse, and divorce rates are no higher than the general population.¹⁵³ In a recent report from Norway, excess suicides were witnessed compared with the general population in patients who received platinum-based chemotherapy.¹⁵⁴

Second Malignant Neoplasms

Both radiotherapy and chemotherapy increase the risk of SMN later in life. Second malignancies can occur 20 years after treatment or later. An exception is acute leukemia, which occurs 2 years to 4 years after chemotherapy. The risk of treatment-related leukemia is proportional to the cumulative dose of etoposide and is estimated to be less than 0.5% for two courses (1,000 mg/m²), less than 1% for three to four courses (1,500–2,000 mg/m²), and as high as 6% for cumulative etoposide doses greater than 3,000 mg/m². Chromosomal translocations involving 11q are characteristic of etoposide-related acute leukemia.¹⁵⁵

Acute leukemia is also seen in a small percentage of patients with mediastinal NSGCT. Although chemotherapy may be a contributing factor, the leukemia associated with mediastinal primary tumors has a separate cause. Megakaryocytic leukemia has been described and may be more common in this setting.¹⁵⁶ Studies have identified the clonal marker *i*(12p) in leukemic cells, indicating that they are clonally descended from the primary germ cell tumor.¹⁵⁷ One hypothesis is that yolk sac tumor (a common histology among mediastinal germ cell tumors) retains the pluripotency

of the normal yolk sac, which functions as a hematopoietic organ during embryogenesis.¹⁵⁸ This association has not been described in yolk sac tumor of testicular origin.

Second primary germ cell tumor of the contralateral testicle occurs in approximately 2% of survivors and occasionally as a synchronous (bilateral) presentation. The risk of second germ cell tumor is present over the lifetime of the individual, making it especially important that the patient is counseled to report any testicular symptoms and that long-term follow-up includes surveillance of the contralateral testicle. Prophylactic radiation of the contralateral testicle has been promoted in Europe as a means to reduce the risk, whereas surveillance alone is standard in the United States. Although adjuvant carboplatin for stage I seminoma appeared to reduce the incidence of second primary tumors, a longer follow-up period is needed to determine whether it is merely a delay in clinical presentation.⁴⁸

LONG-TERM FOLLOW-UP

The mandatory duration of follow-up for detection and management of recurrence of germ cell tumors is 5 years, though late relapses are known to occur, and survivorship programs remain an important tool for these patients. There is wide consensus, however, that testicular cancer survivors should have lifelong follow-up, whether it is at the primary treatment center or with a general internist who is knowledgeable about survivorship issues. Long-term follow-up is necessary for detection of late recurrences, second testicular primaries, and SMN. Survivors require management of cardiovascular effects and symptoms of neurotoxicity. Special care may be required for maintenance of sexual health, fertility issues, and psychosocial functioning.

MIDLINE TUMORS OF UNCERTAIN HISTOGENESIS

Tumors of unknown primary site in young patients occasionally respond well to cisplatin-based chemotherapy. This can be the presentation of an extragonadal germ cell tumor. The diagnosis should be considered in relatively young patients in whom the tumor has predominantly midline distribution and histologic appearance of poorly differentiated carcinoma. Serum tumor markers may or may not be elevated. Molecular cytogenetic analysis for 12p genetic content can help to confirm the diagnosis but is often inconclusive. Immunohistochemical markers SALL4, OCT3/4, CD117, SOX2, CD30, and low-molecular-weight keratins can also facilitate the diagnosis.

OTHER TESTICULAR TUMORS

Sex Cord and Gonadal Stromal Tumors

Leydig and Sertoli Cell Tumors

The most common sex cord stromal tumors in men are Leydig cell and Sertoli cell tumors (Table 47.6). These tumors are occasionally metastatic, but the majority are benign. Treatment is radical orchiectomy. The risk of metastasis has been associated with vascular invasion, cellular atypia, tumor necrosis, infiltrative margins, increased mitotic rate, tumor size greater than 5 cm, older age at presentation, increased proliferation index, and aneuploidy. The pattern of metastasis is initially to the retroperitoneal lymph nodes.^{159,160}

Leydig and Sertoli cell tumors are associated with steroid hormone hypersecretion. Sertoli cell tumors may be accompanied by precocious puberty in boys. Patients with Leydig cell tumors may have decreased libido and gynecomastia or virilization in prepubertal boys.

TABLE 47.6

World Health Organization Histologic Classification of Testicular Sex Cord and Gonadal Stromal Tumors

Leydig cell tumor
Malignant Leydig cell tumor
Sertoli cell tumor
Sertoli cell tumor lipid-rich variant
Sclerosing Sertoli cell tumor
Large-cell calcifying Sertoli cell tumor
Malignant Sertoli cell tumor
Granulosa cell tumor
Adult-type granulosa cell tumor
Juvenile-type granulosa cell tumor
Tumors of the thecoma or fibroma group
Thecoma
Fibroma
Sex cord or gonadal stromal tumor: incompletely differentiated
Sex cord or gonadal stromal tumors, mixed forms
Malignant sex cord or gonadal stromal tumors
Tumors containing both germ cell and sex cord and gonadal stromal elements
Gonadoblastoma
Germ cell–sex cord/gonadal stromal tumor, unclassified

With permission from Eble JN, Sauter G, Epstein JI, et al., eds. *World Health Organization Classification of Tumours. Pathology and Genetics of Tumours of the Urinary System and Male Genital Organs*. IARC Press; 2004.

Chemotherapy and radiotherapy are not known to be effective for metastatic sex cord stromal tumors. There are several reports of treatment with mitotane, which was used because it is known to be effective in adrenocortical carcinoma, another steroid-producing tumor. Reported results with mitotane treatment of metastatic sex cord stromal tumors have been mixed, but it can be considered, especially for functional tumors with symptom of steroid hormone excess.

Granulosa Cell Tumors

Granulosa cell tumors of the testicle are rare.¹⁶¹ They resemble granulosa cell tumors of the ovary. Tumors secrete estrogen, and patients may present with gynecomastia. Treatment is radical orchiectomy, which is usually curative. Granulosa cell tumor (juvenile type) is the most common testicular neoplasm in neonates.

Gonadoblastoma

Gonadoblastoma contains both germ cell and sex cord stromal elements.⁹ It is associated with testicular dysgenesis and karyotypic anomalies. It has potential for metastasis of the germ cell component of the primary tumor.

Mesothelioma

Mesothelioma of the tunica vaginalis can invade the testis and spermatic cord.¹⁶² Treatment is radical orchiectomy with complete excision of the spermatic cord and hemiscrotum. RPLND should be considered in patients with LVI or invasion of the testicular parenchyma.

Adenocarcinoma of the Rete Testis

Adenocarcinoma of the rete testis has a poor prognosis.¹⁶³ It is not responsive to radiotherapy or chemotherapy. Treatment is radical orchiectomy, and 30% to 50% of patients die within 1 year of metastatic disease. RPLND should be considered for selected patients and may be curative for patients with low-volume metastatic disease.

Epidermoid Cyst

Epidermoid cyst is often asymptomatic.¹⁶⁴ On palpation, it is firm and sharply demarcated, and it appears cystic on ultrasound examination. It is of uncertain relation to germ cell tumors and is not associated with ITGCN. Histologically, the cyst is lined with squamous epithelium, and the adjacent testicular parenchyma is benign. The clinical course is benign. Treatment with enucleation of the tumor or radical orchiectomy is curative.

Lymphoma

Lymphoma presents as painless enlargement of the testicle and may be bilateral.⁹ It is the most common secondary malignancy of the testis in men older than 50 years. It typically occurs in the setting of advanced systemic disease, often accompanied by central nervous system or bone marrow involvement.

Metastatic Carcinoma

Metastatic carcinoma is rarely confused with primary testicular cancer. It occurs most commonly in the setting of advanced disseminated disease. Treatment is determined by the type of primary tumor.

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